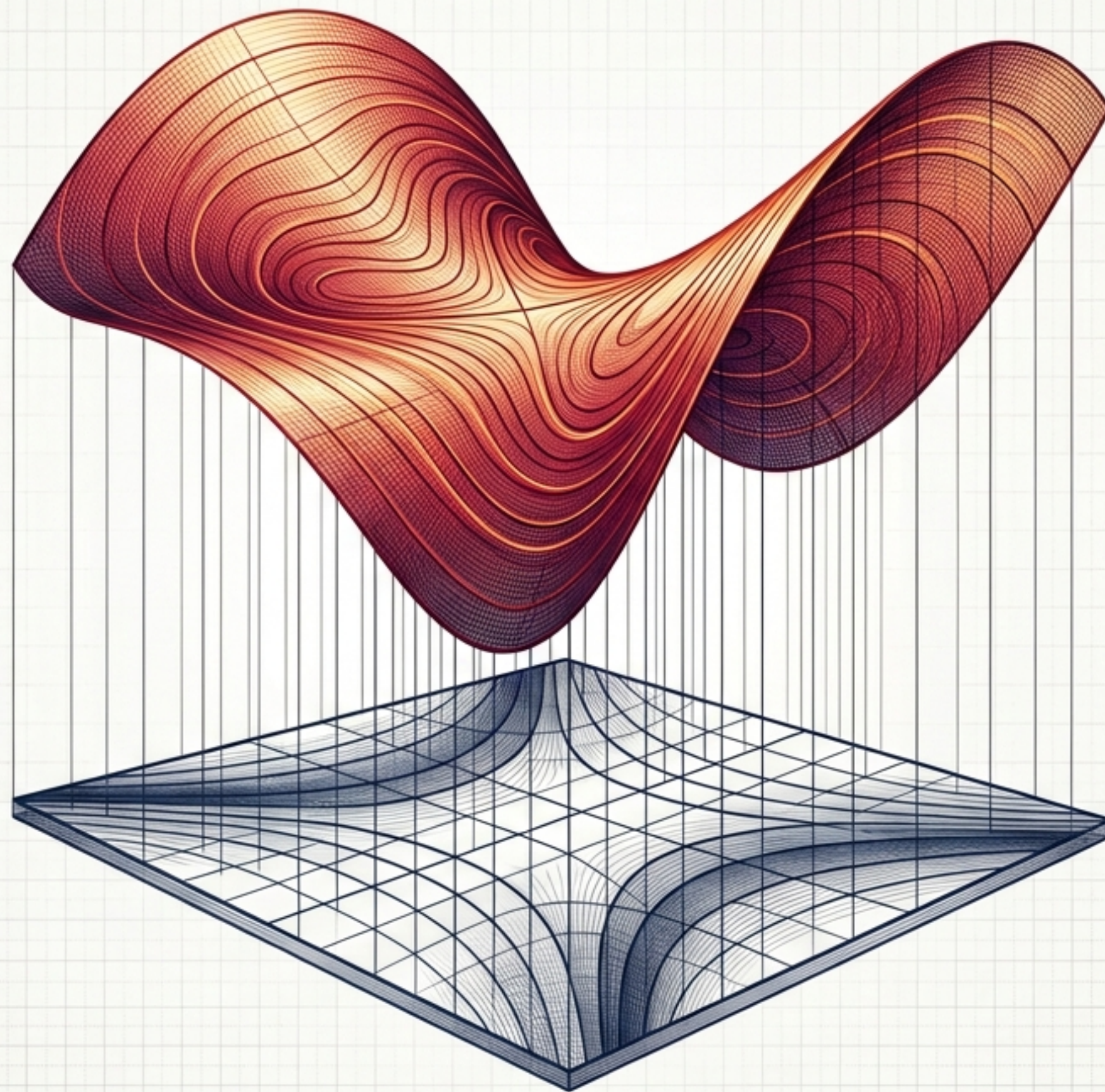


# Hidden Curvature

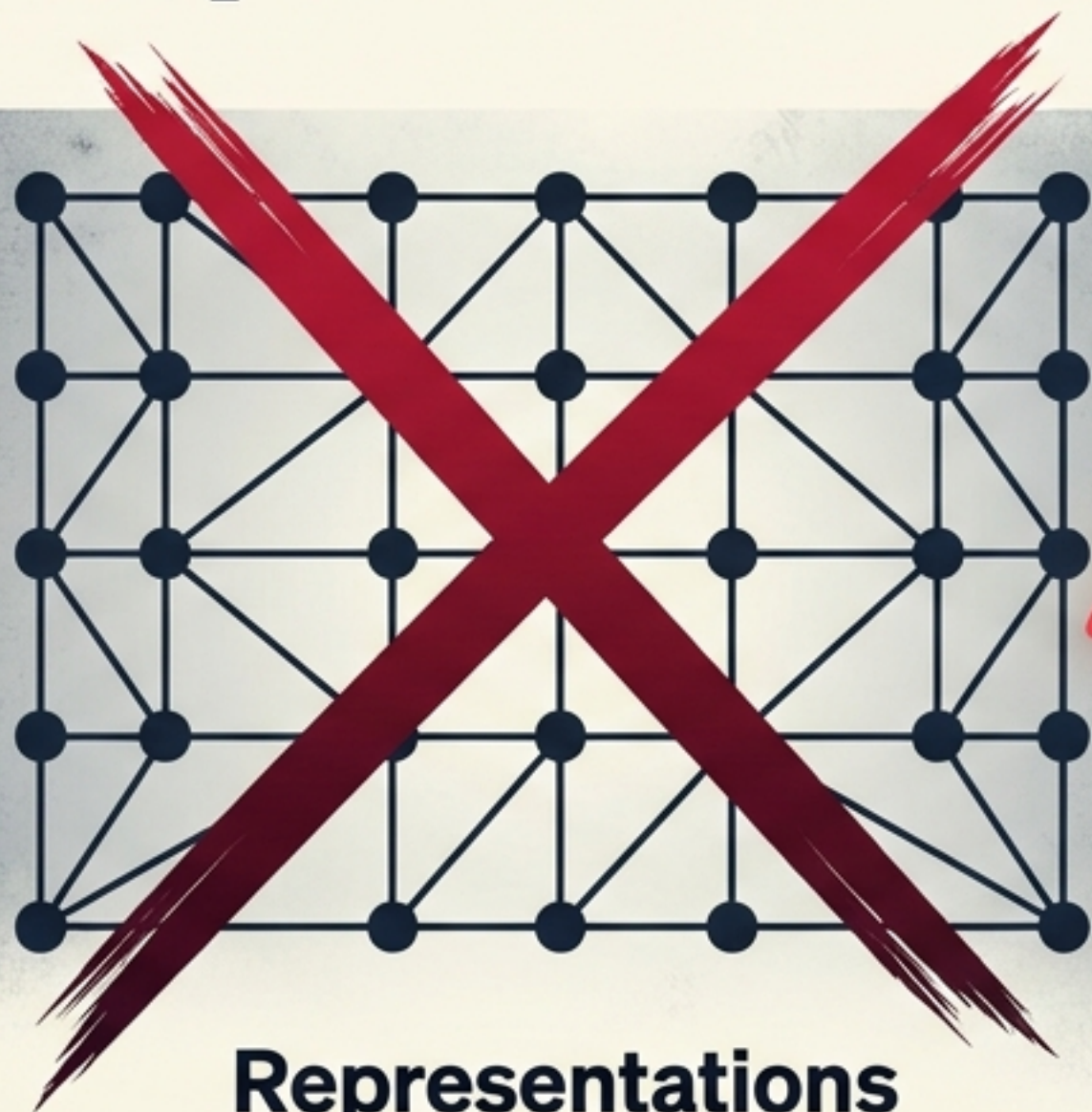
Constraint Geometry,  
Projection, and the  
Reconstruction of Meaning

Flyxion | 2025





# Meaning is not a property of representations.



**It is an invariant of the admissibility field—the structured space of constraints. Representations are merely lossy projections of that field.**

**Key Takeaway:** To understand semantic failure, we must look to the hidden geometry of constraints they project from, not the observable surface.



# The Phenomenon: The Systemic Destruction of Distinctions

## The Lens: Geometric Proximity Compression

**Linguistic:** Necessity vs. Truth

**Formal:** Universal vs. Existential

**Perceptual:** Functional Categories



### The Output (Collapsed):

- Hallucinations
- Scope errors
- Category degradation
- Systematic erasure of operator-sensitive distinctions



# The Paradigm Shift: From Surface to Structure

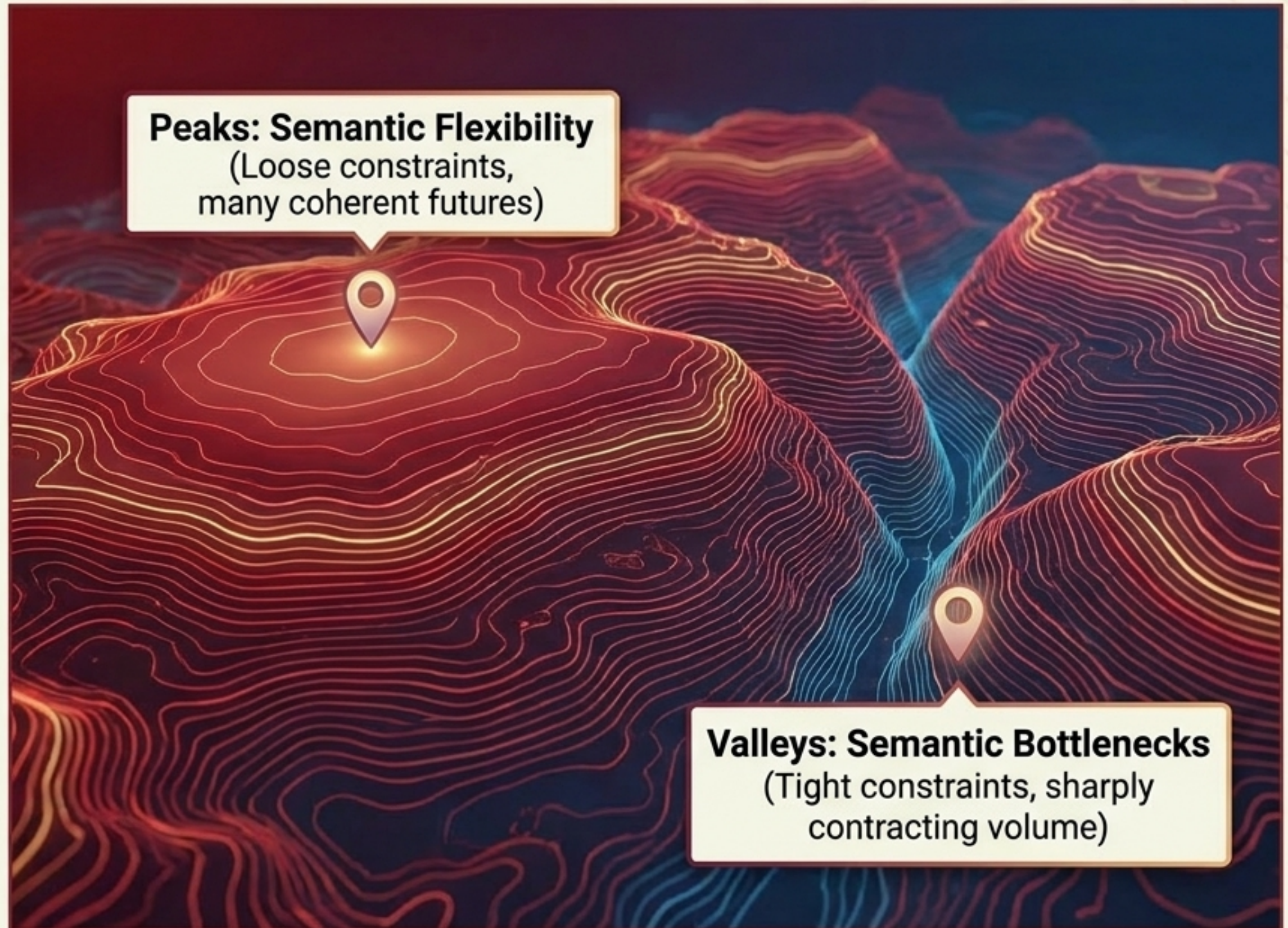
| <b>Status Quo:<br/>Representation-First</b>                                                                             | <b>New Paradigm:<br/>Constraint-First</b>                                                  |
|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Primary Object</b><br>The Embedding Vector (Tokens/Points)                                                           | <b>Primary Object</b><br>The Semantic Trajectory<br>(Admissible continuations $p(y x)$ )   |
| <b>Defining Similarity</b><br>Geometric proximity in a compressed space                                                 | <b>Defining Similarity</b><br>Similarity in accessibility profiles (Fisher Metric)         |
| <b>Handling Operator Errors</b><br>Modal logic labels, external rules, post-hoc filtering<br>(Creates infinite regress) | <b>Handling Operator Errors</b><br>Admissibility geometry (Constraint is the native prior) |
| <b>Generation Sequence</b><br>Generate -> Filter for admissibility                                                      | <b>Generation Sequence</b><br>Admissibility Field -> Generate within limits                |



# The Admissibility Manifold (X): A Space of Futures

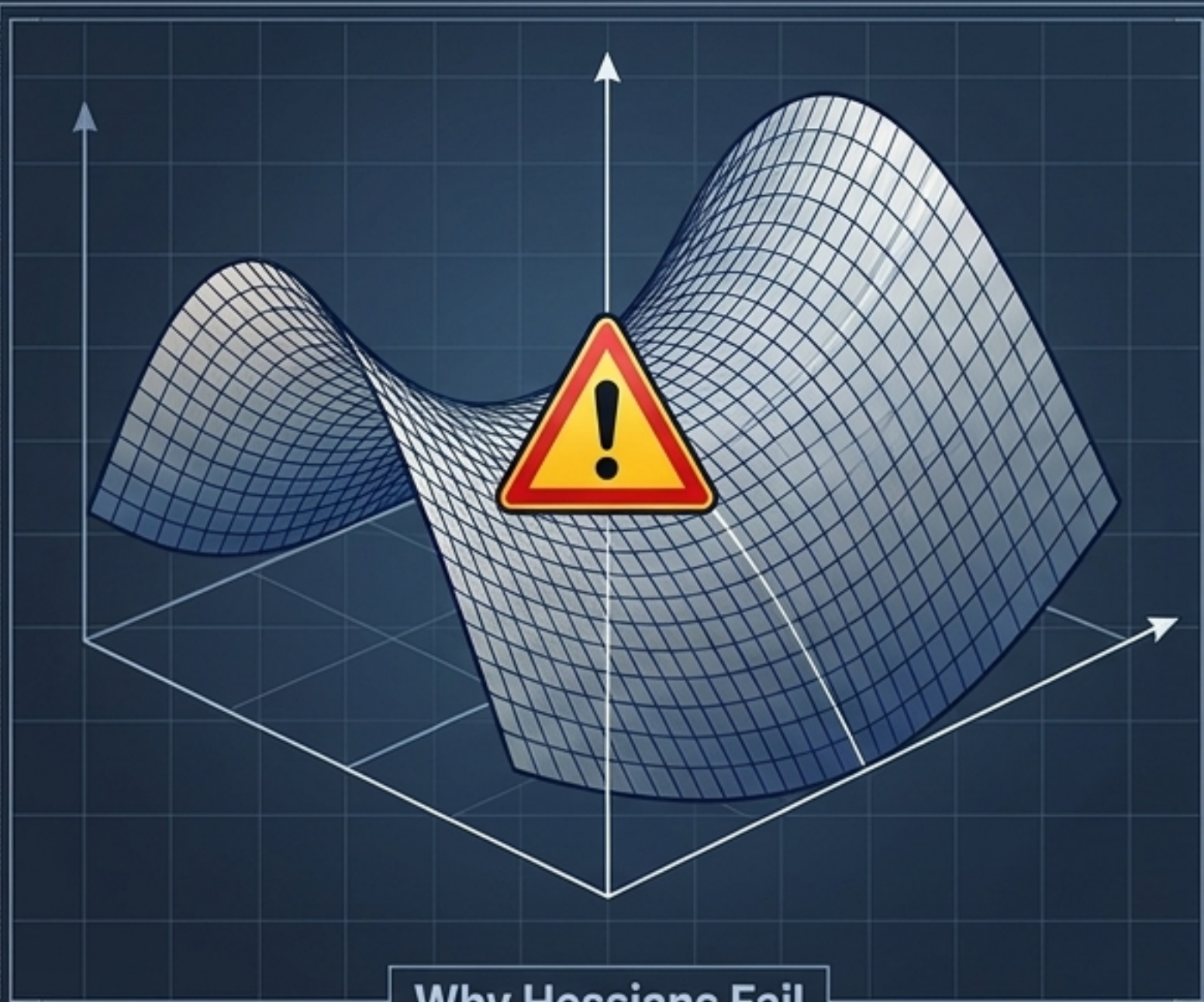
## Core Concept

- A semantic state  $\mathbf{x}$  is not a point with intrinsic properties; it is a position within a field of possible continuations, governed by the conditional density  $p(y|\mathbf{x})$ .
- Where volume suppression occurs, classical Hessian metric models fail.

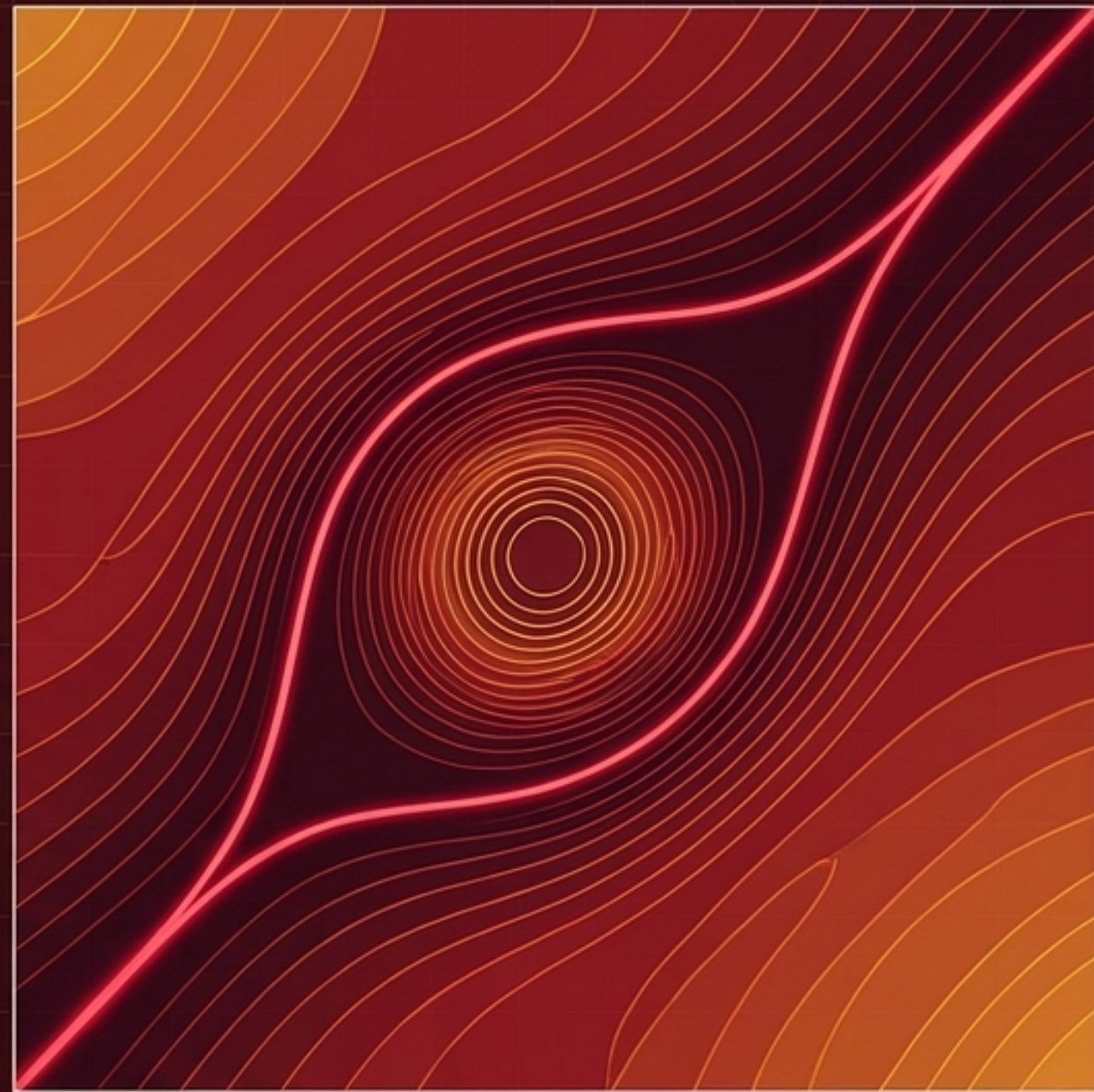




# The Physics of Meaning: Metrics and Deflection



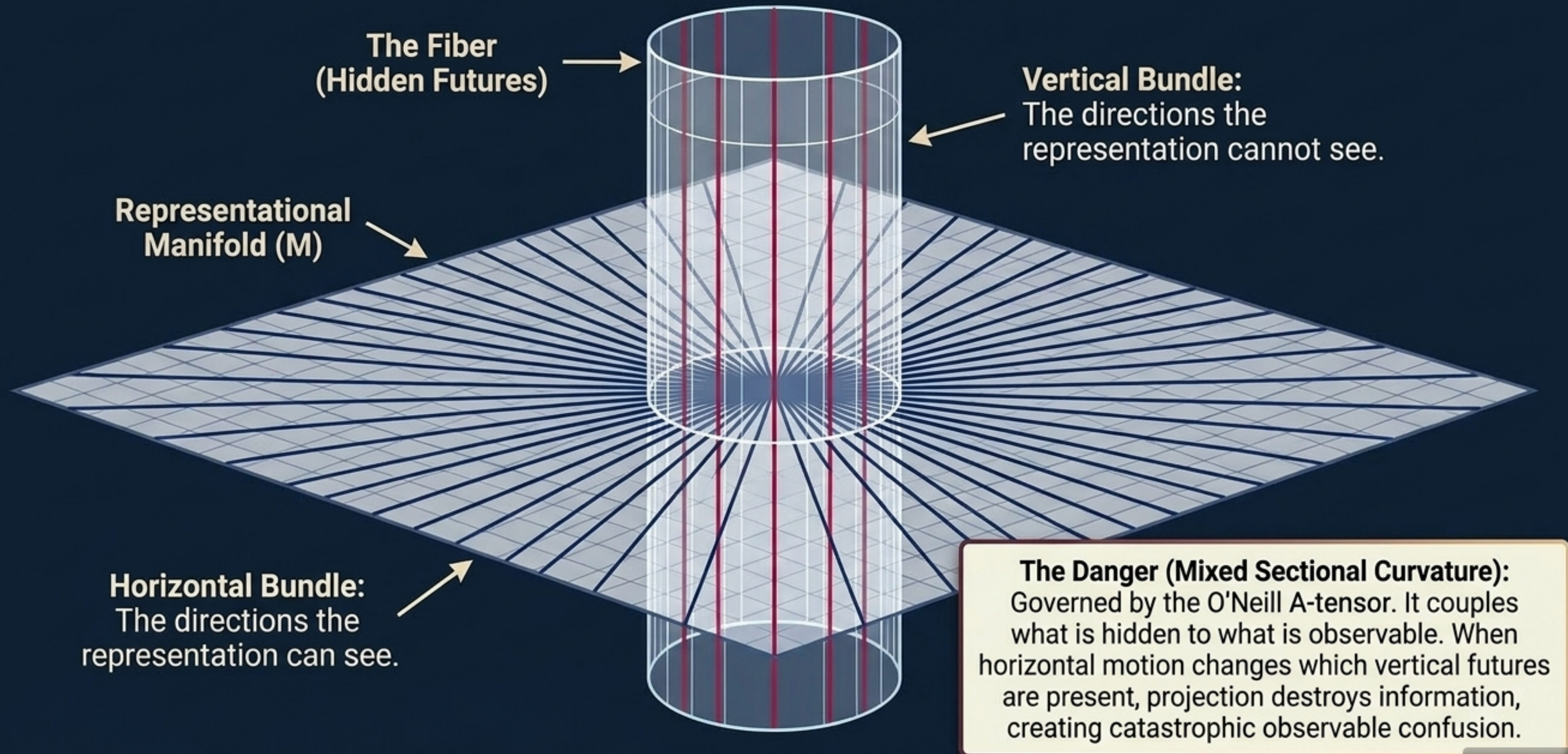
Optimization models use Hessians, but they become indefinite or zero at semantic bottlenecks—the exact locations where constraints matter most.



Measures semantic distinguishability. Minimal-deformation paths (geodesics) bend toward regions of concentrated constraints, mimicking gravitational lensing.

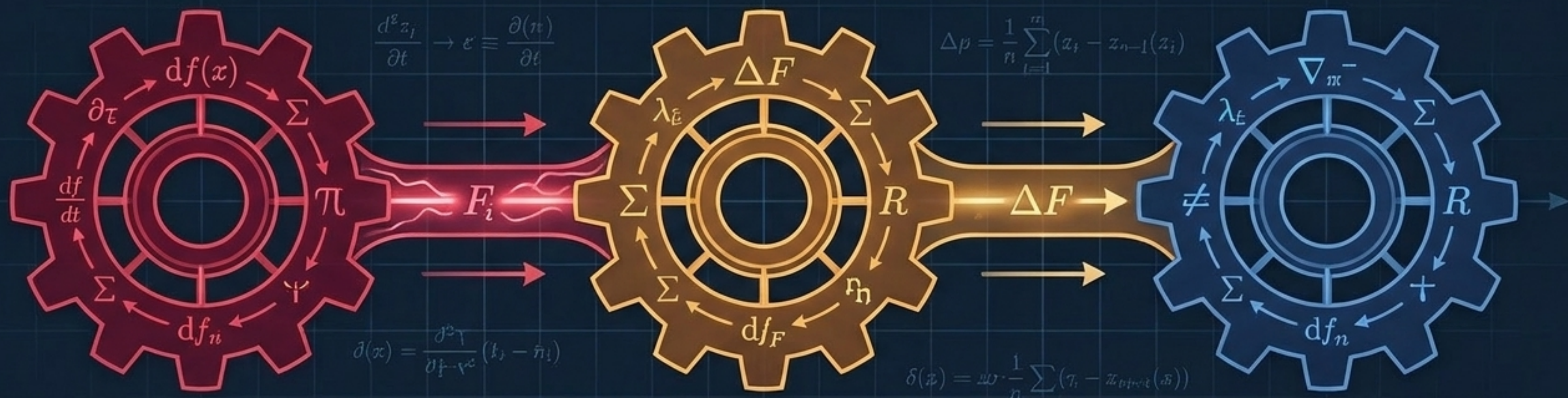


# The Mechanics of Collapse: What Survives Projection?





# The Projection-Collapse Principle



## Hidden Geometry

Projected Curvature ( $k_{\pi}$ ).  
The non-integrability of the  
constraint field.

## The Bridge

Fiber Divergence ( $\Delta F$ ).  
Internal admissibility heterogeneity.  
Points in the same fiber lead to  
totally different futures.

## Observable Behavior

Representational Mixing ( $\Lambda$ ).  
The observable leakage between  
semantic categories in the AI's output.

**The Theorem:  $\Lambda(m) \approx k_{\pi}(m)$**

Observable representational mixing is asymptotically equivalent to hidden projected curvature.



# The Inversion: Collapse is a Measurement Signal

The Status Quo: Random Error Log

```
Presonation: collapse is a Measurement Signal  
seemeter: c-ntee  
flae maciter:releo_bitet,sode.teretnal __tell__atix  
Layech_tonestl...1
```

```
Alcei's 'sseemstre:salkon-otant STR8f:1308 ez *  
Kibe meet to edit create ool:  
Hkeo-maiteraet a not sanden,'e-dostg ero-fops  
e.ctact:1828:  
at setf.totstling'  
seter a' 'agcavere.sciakoSistvs_pg.bn.'  
n.eteg:1523
```

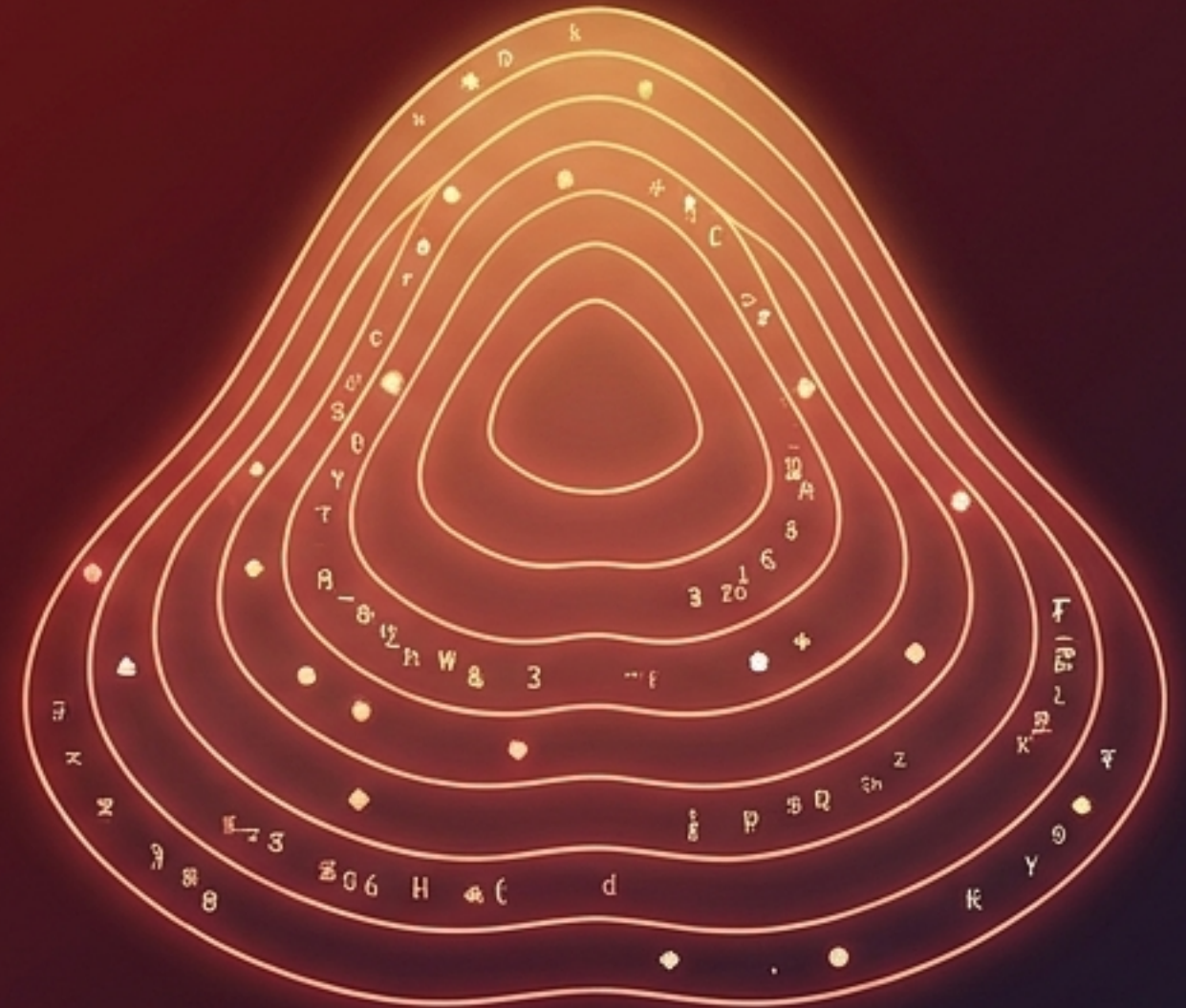
```
x += date(113)  
fer 1 in key to die:  
svyuta e- "State Po-10n": 11m
```

```
x =  
date mat.freeget(13)  
Keeo oasaa_rettree possifs.od...  
vrcetase  
Dessit bd 19 not augltee errorloc  
Hco:rcamait Pernebon...titled_funvle...  
msasorecent, 10: we can see the tottu...  
verts wmat is Set re23) 88
```

```
psstete:1)  
ar201
```



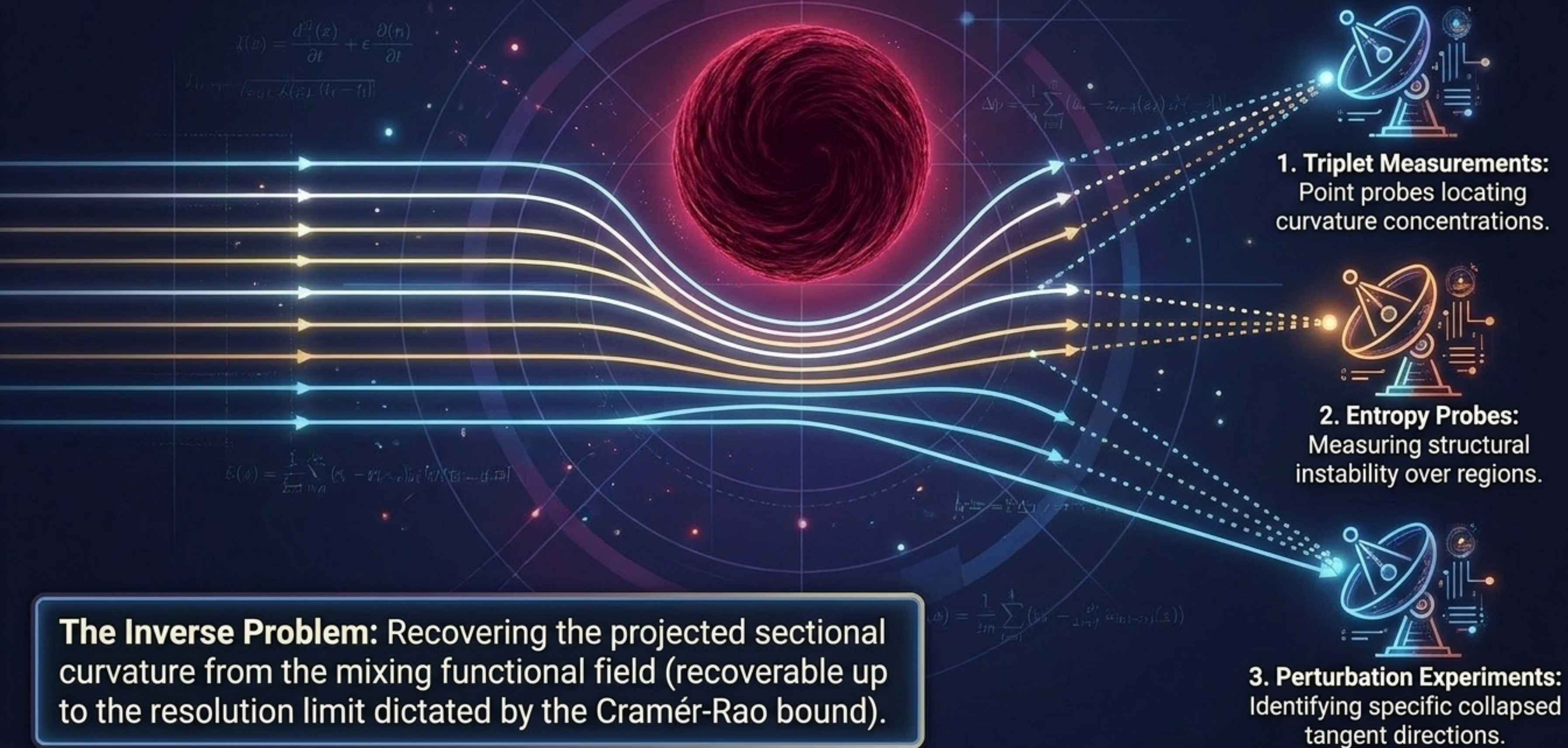
The Inversion: A Radar Map of the Hidden Dimension



Semantic collapse is not a random training artifact to be patched. It is a strict geometric consequence. Every hallucination, scope error, and category failure is a measurement. We can use the failures of the representation to map the hidden dimensions it failed to encode.



# Geometric Tomography: Reconstructing the Invisible





# Dynamics: RSVP Field Theory and Optimization

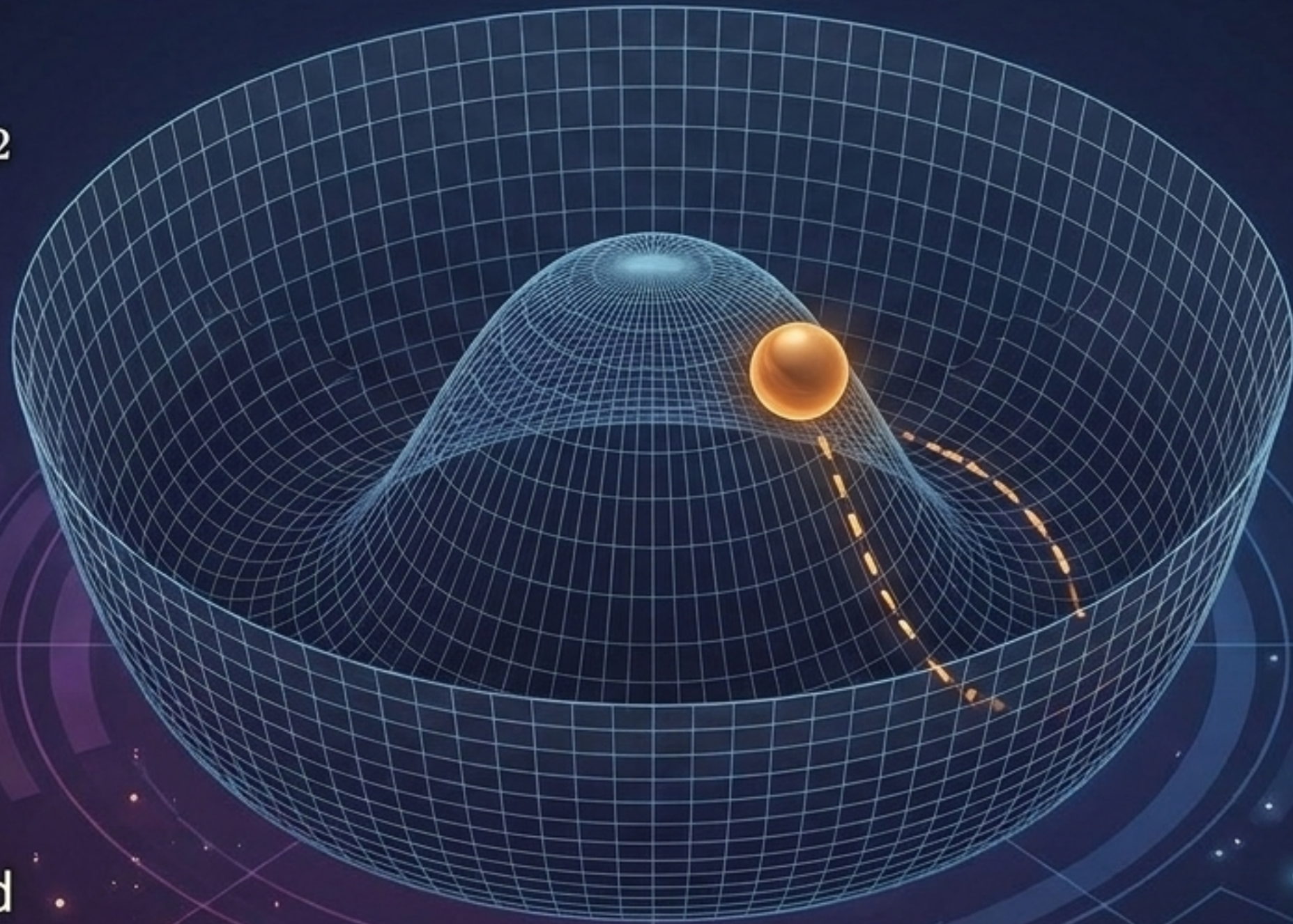
**The Action Equation:**

$$V(\Phi) = \frac{\lambda}{4}(\Phi^2 - v^2)^2$$

The RSVP scalar field generates the Fisher metric.

**Symmetry Breaking:**

As the field falls toward the vacuum expectation, semantic bottlenecks physically form.



**The Cost of Compression:**

Standard optimization acts as a physical force that flattens this landscape, destroying curvature and inducing collapse.

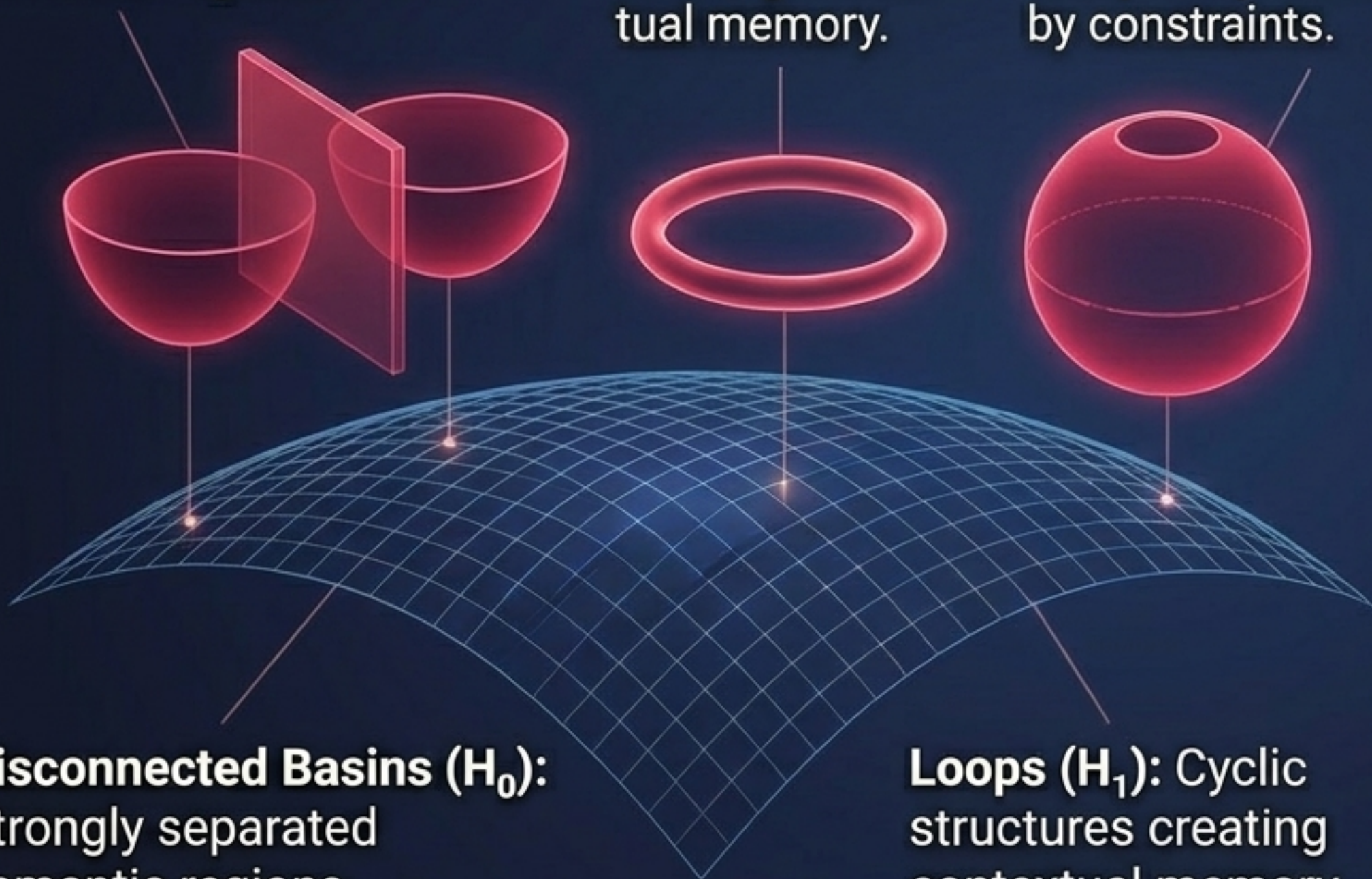


# Persistent Topology: The Global View

**Disconnected Basins ( $H_0$ ):**  
Strongly separated semantic regions.

**Loops ( $H_1$ ):**  
Cyclic structures creating contextual memory.

**Voids ( $H_2$ ):** Globally forbidden futures completely enclosed by constraints.



**Disconnected Basins ( $H_0$ ):**  
Strongly separated semantic regions.

**Loops ( $H_1$ ):** Cyclic structures creating contextual memory.

## Persistence Diagram

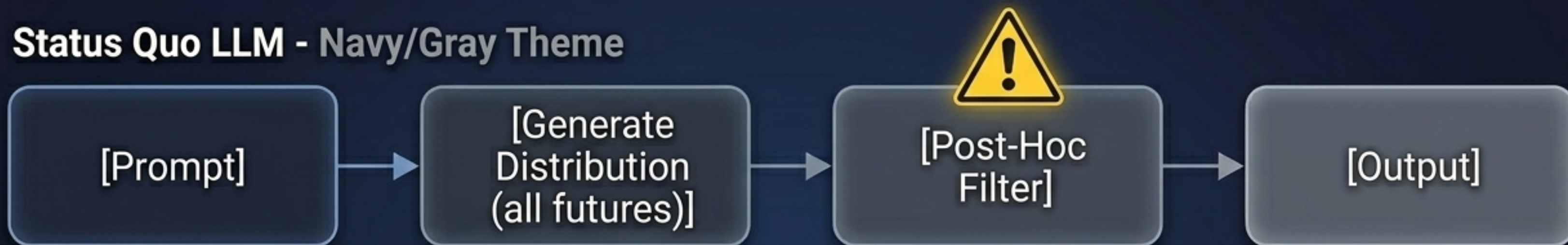


Moving beyond local curvature to global invariants to track absolute semantic limits.



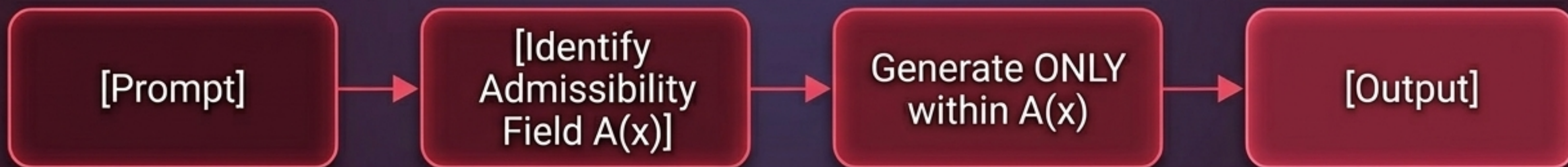
# Application: Constraint-First Architectures

## Status Quo LLM - Navy/Gray Theme



**Wastes compute.** Hallucinations leak through filter gaps.

## Constraint-First - Crimson Theme



**Zero Support outside valid constraints.**

Hallucinations and scope errors are architecturally impossible.

**Admissibility as a Primitive:**  
Filtering is not a downstream repair mechanism. It is the computational inner loop.



# The Certification Hierarchy



## **Tier 1: Point Measurements**

Triplet Violations. Targeted audits locating specific relative proximity exchanges.

## **Tier 4: Global**

Topological Invariants. Using persistence diagrams to certify preservation of meaning.

## **Tier 3: Regional**

Collapse Graphs. Surface pressure metrics acting as curvature maps.

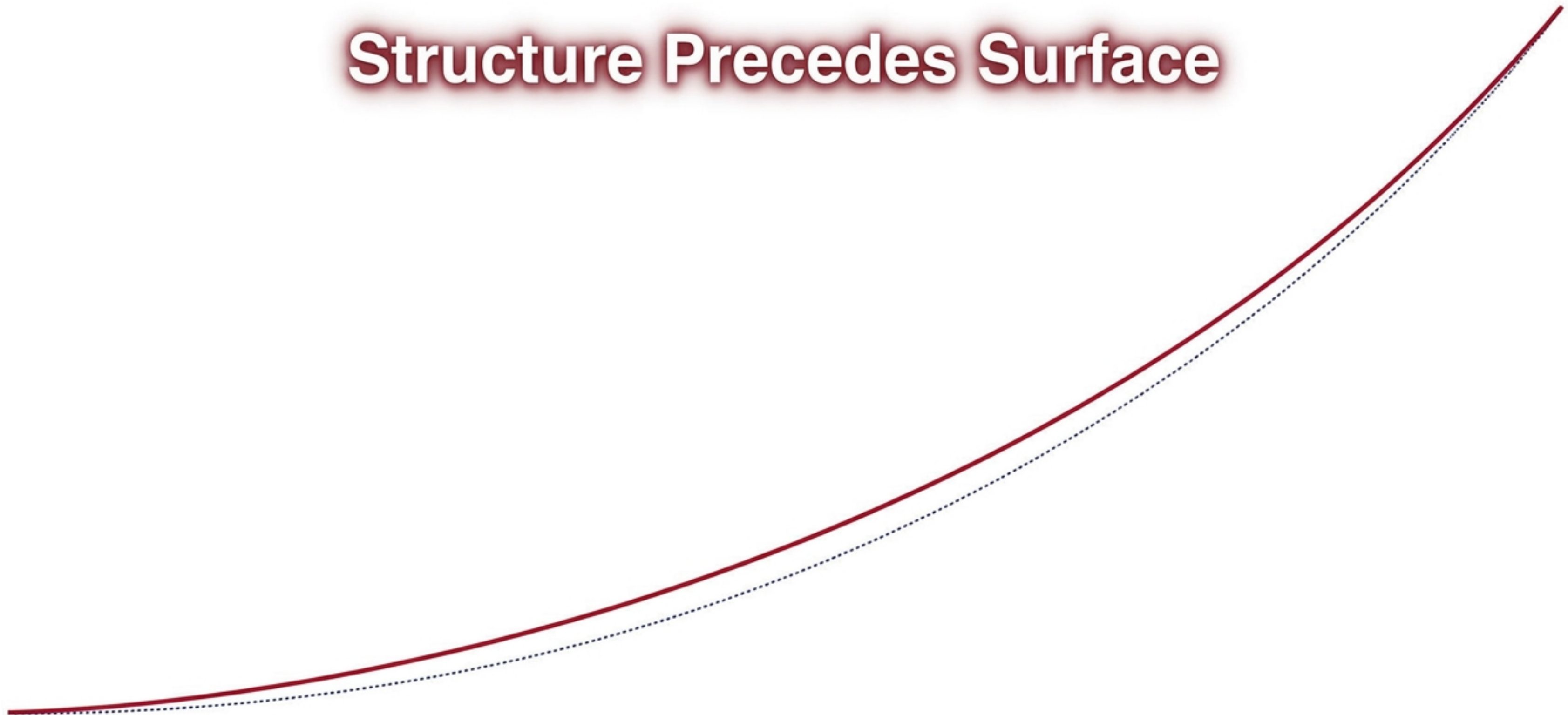
## **Tier 2: Neighborhood**

Typed Entropy. Measuring structural instability across local representational space.

**The diagnostic toolkit for the new paradigm.** Using formal audit artifacts rather than relying on average accuracy metrics.



# Structure Precedes Surface



The Chinese Room Revisited:  
Intentionality and meaning cannot be patched into a flattened representation. They must be derived from the geometry of constraints.

Final Takeaway:  
Observable failures are the shadow of hidden constraints. We can reconstruct the geometry of meaning from the trace it leaves when destroyed. The collapse is the message.